

CHEM 427/527 · SESSION 4 · TUESDAY, OCTOBER 13, 2026

Chemical data, formats, and structures

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Read a JCAMP-DX file as a document: name its header fields, its data block, and the five fields that fix what the numbers mean.
2. Parse the header into a dictionary with `open()`, string slicing and `.get()`, and expand the data block into wavenumber and absorbance arrays.
3. Check a parsed spectrum against the file's own consistency fields, and say which check catches which reading mistake.
4. Name what the file records and what it does not, and the question a collaborator could not answer without it.

FIRST, EVERYONE PULLS

Get session 4 with git pull

Bring in session 4

```
git pull
```

A new folder appears:

```
S04-chemical-data-formats
```

Laptop still failing?

- Work beside a neighbor, on their laptop, today.
- See Lucas or Chris in the third hour.
- Submit your own notebook from a working laptop by Thursday.

Submissions are individual from today.

WHERE SESSION 2 STOPPED

What the CSV never said

ms_peaks.csv

```
mz,intensity_counts  
130.8413,11148  
138.0663,190114  
144.2931,9449
```

What did you have to know to use it
that the file never said?

1. Values

The numbers. A CSV carries this layer by default.

2. Acquisition metadata

Instrument, settings and units.

3. Provenance

Who measured what, when, and from which sample.

Today's file carries the first two layers and part of the third.

A FILE READ AS A DOCUMENT

Acetone's header, line by line

data/acetone_ir.jdx, lines 1 to 12

```
###TITLE=Acetone
###JCAMP-DX=4.24
###DATA TYPE=INFRARED SPECTRUM
###ORIGIN=Sadtler Research Labs Under US-EPA Contract
###OWNER=NIST Standard Reference Data Program
Collection (C) 2018 copyright by the U.S. Secretary of Commerce
on behalf of the United States of America. All rights reserved.
###CAS REGISTRY NO=67-64-1
###MOLFORM=C 3 H 6 O
###$NIST SOURCE=MSDC-IR
###STATE=gas
###XUNITS=1/CM
```

One field per line, as
##NAME=value: 23 fields in
25 lines.

Two lines continue
##OWNER and start without
##.

After ##XYDATA come 88
data lines, then ##END=.

PARSING IT, LIVE

A line, then a dictionary

A header line

```
line = "###XUNITS=1/CM"
line[2:]
'XUNITS=1/CM'
line[2:].split("=", 1)
['XUNITS', '1/CM']
```

A dictionary

```
header = {"XUNITS": "1/CM",
          "DELTAX": "4.0"}

header["DELTAX"]
'4.0'

header.get("RESOLUTION")
None

header["RESOLUTION"]
KeyError: 'RESOLUTION'
```

Every value is text until float() converts it.

THE DATA BLOCK

The first data line, read aloud

450.0 40 76 71 59 67 74 97 132 180 202

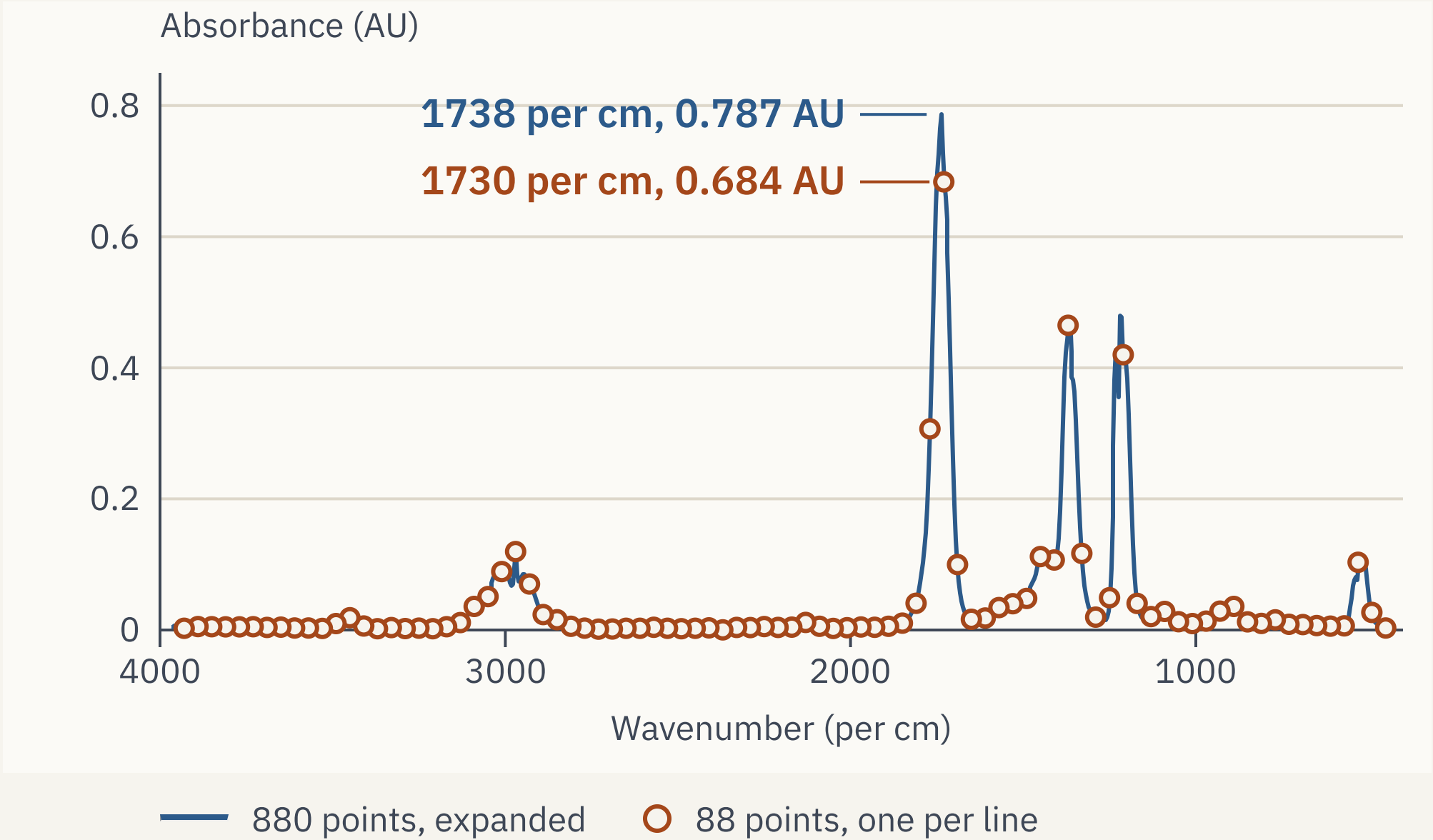
Wavenumber	450	454	458	462	466	470	474	478	482	486
Stored integer	40	76	71	59	67	74	97	132	180	202

Header field	What it says about this line
##XYDATA=(X++(Y..Y))	One x value, then the y values from that x on
##DELTAX=4.0	x steps by 4 per cm: 450, 454, ... 486
##YFACTOR=0.000078659	Integer times factor: 40 × 0.000078659 = 0.003146 AU, the header's ##FIRSTY

A line is not a point, and the integers are not absorbances.

THE READING THAT LOOKS RIGHT

Same shape, wrong numbers



Field	Header	Per line
NPOINTS	880	88
LASTX	3966	3930
MAXY	0.787	0.684

One point per line fails all three, and moves the strongest band by 8 per cm.

The plot looks like a spectrum. The header's three numbers are the test.

WHAT 8 PER CM COSTS

A genuine batch fails

1.0 per cm

The European Pharmacopoeia's tolerance on an FT-IR wavenumber scale.

8 per cm

How far one point per line moves acetone's carbonyl band.

To accept a batch as the compound on its label, an analyst compares its infrared bands with a reference spectrum. A reference read one point per line fails a genuine batch.

YOUR MOLECULE, IN SECTION 2

Download your record once

1. Set MY_CAS in the Parameters cell to your CAS number.
2. Run the cell that prints the WebBook address, and open it in your browser.
3. Drag the downloaded file onto the data folder in the VS Code Explorer.
4. Keep the name, for example 64-17-5-IR.jdx. A second copy, with (1) in its name, breaks grading: check it against what the cell printed.

Take turns, by table

The WebBook refused about 40 requests in 20 seconds from one address. This room shares one address.

Three records are Coblenz transmittance spectra with a different DELTAX. That is on purpose: section 2 says what to do.

ASSIGNMENTS

One molecule each

#	Molecule	CAS
1	ethanol	64-17-5
2	methane	74-82-8
3	2-propanol	67-63-0
4	acetonitrile	75-05-8
5	water	7732-18-5
6	dichloromethane	75-09-2
7	carbon tetrachloride	56-23-5
8	cyclohexane	110-82-7
9	benzene	71-43-2

#	Molecule	CAS
10	toluene	108-88-3
11	chlorobenzene	108-90-7
12	aniline	62-53-3
13	phenol	108-95-2
14	tetrahydrofuran	109-99-9
15	dimethyl ether	115-10-6
16	ethyl acetate	141-78-6
17	acetic acid	64-19-7
18	formic acid	64-18-6

#	Molecule	CAS
19	nitrous oxide	10024-97-2
20	cyclohexanone	108-94-1
21	acetophenone	98-86-2
22	carbon dioxide	124-38-9
23	dimethyl sulfoxide	67-68-5
24	diethyl ether *	60-29-7
25	formaldehyde *	50-00-0
26	acetaldehyde *	75-07-0

* Coblentz Society record: transmittance, with a different DELTAX. Section 2 says what to do.

AI PRACTICE 1 OF 8: ASK QUESTIONS AS YOU WOULD A SENIOR COLLEAGUE

Ask before you change anything

Ask, in Claude Code

How would you find the strongest band in @data/acetone_ir.jdx, and which line of its header would change your answer?

Read the answer against

```
###YUNITS=ABSORBANCE
```

Then ask

What in my file told you that?

Twice in the notebook: before TASK 3, read against YUNITS, and before section 3, read against NPOINTS and MAXY.

Your band, and one entry

(a) Your answer

- The wavenumber and value of your strongest band
- The header field you needed for each
- What you would have reported without it

(b) One judgment entry

- What the assistant suggested
- What you did: changed, rejected or verified
- Why, from your own file

Scored 0 to 2 from today. You write it yourself.

SECTION 1 IS READ AND RUN. SECTION 2 IS YOURS.

Go, then submit

Submit

Restart, Run All, save, and upload activity.ipynb by **1:30 pm Thursday, October 15.**

Keep your file in data/ under its WebBook name. The graders run your notebook against it.

Scoring, 0 to 3

1 if it runs after a restart, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas or Chris in the room.

The third hour is an office hour, first for laptops that still fail.